

ADITI ANAND MUPPAVARAM

Business Analyst | Data Analyst — SQL · Process Automation · Product Requirements

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PROFESSIONAL SUMMARY

Business Analyst / Data Analyst with 3+ years driving data-driven process automation, reporting standardization and product enhancements that improve operational efficiency for large-scale CMS federal health IT systems. Partners with technical and product stakeholders to define requirements, prioritize features and influence product roadmaps; applies systems-level thinking to assess tooling capabilities and gaps in backend data workflows. Writes, optimizes and maintains SQL queries across large-scale operational datasets and builds the query logic, data models and backend logic (stored procedures and functions) behind dashboards, tracking tools and automated workflows — enabling scalable, standardized operations. Delivered a self-service dashboard that improved data visualization 60%, 50+ KPI reports, process improvements that cut data processing time 40%, and AI/ML-based predictive models at 85% accuracy on cross-functional projects with multiple stakeholders.

SKILLS

SQL & Database Systems: SQL query writing, optimization and maintenance; query migration and validation across data-model changes; PL/SQL stored procedures and functions (backend logic); MySQL, PostgreSQL, Oracle, MongoDB, AWS DynamoDB; data modeling; database administration (users, roles, security patches)

Data Pipelines & Analytics: Data pipeline design, ETL pipelines, data warehouses; Python (pandas, NumPy), R; statistical and quantitative analysis; A/B testing; AI/ML-based predictive modeling (logistic regression, decision trees, random forests)

Dashboards & Reporting: Tableau, Power BI, Excel; KPI dashboards and tracking tools; operational reporting; self-service enablement (dashboards, reports, operational support documentation); data-driven decision-making

Large-Scale Automation & Workflow Management: Process automation across large-scale operational datasets (PL/SQL automation of daily data operations, automated workflows), Selenium test automation, workflow analysis and redesign, process mapping (Visio), gap analysis, operational standardization, UAT and regression testing

Business Analysis & Product: Requirements elicitation and definition, BRDs, functional specifications, user stories and acceptance criteria, feature prioritization, product roadmaps, backlog management, wireframing, stakeholder management

Project Management & Agile: Cross-functional project management, project timelines, deliverables, status reporting, stakeholder communications, Scrum, SAFe (PI planning), Jira and Confluence (ticketing and workflow tools)

PROFESSIONAL EXPERIENCE

Tria Federal (formerly Softrams) — Windsor Mill, MD · CMS federal health IT programs 07/2023 – Present

Product Analyst — CMS MEARIS 10/2024 – Present

CMS's intake and review platform for Medicare payment and coding applications (NTAP, MS-DRG, ICD-10-PCS).

- Partner with technical and product stakeholders across 5+ CMS business units to define requirements, prioritize features and influence the product roadmap; maintain a backlog in which every feature traces to program goals and release commitments.
- Author user stories, acceptance criteria and wireframes that turn operational pain points into product requirements, reducing sprint rework and clarification cycles and improving team productivity.
- Drive roadmap planning and SAFe/Scrum ceremonies (PI planning, sprint planning, backlog refinement); manage project timelines, deliverables and stakeholder communications across internal teams, improving on-time delivery and sprint predictability.
- Streamline and simplify intake and review process workflows with UX/UI from user feedback, raising usability and task-completion rates.
- Plan and execute UAT (manual and regression) against acceptance criteria, tracking defects to resolution before release; support go-live and cutover activities.

CMS ARTS — Analysis, Reporting and Tracking System (tracking tool) for contractor proposals, cost reports, deliverables and workload.

- Ran requirements workshops with CMS stakeholders to align on operational scope and onboard new functions and features; documented BRDs, functional specifications and process maps in Confluence as standardized, self-service reference material.
- Analyzed financial data to design and deliver the Financial Monitoring dashboard — improving data visualization 60% and giving program leads a single, self-service view of contractor spend for decision-making.
- Assessed system gaps and shaped solution designs against business goals (20% increase in project success rate); executed test plans that cut the defect rate 18%; supported project planning, tracking and status reporting.

CMS HPMS — Health Plan Management System for Medicare Advantage and Part D data collection, compliance and reporting.

- Wrote, optimized and maintained SQL and Python queries across large-scale operational datasets (Medicare Advantage and Part D) for reporting and decision-making; unit-tested stored procedures and functions before production deployment to protect data continuity and accuracy.
- Delivered 50+ standardized analytical reports and visualizations tracking program KPIs for CMS stakeholders; drove cross-team process improvements that cut data processing time 40%.
- Collected, cleaned and preprocessed healthcare data from multiple upstream data sources; defined data-design requirements for extensions to existing data models in a long-running legacy system with cross-functional teams, aligning cross-team data and system dependencies and supporting query migration as models changed.

Job Ring — Employment portal: production analytics, predictive modeling and data operations.

- Led process automation initiatives for daily data operations with reusable SQL procedures (PL/SQL) that run as automated workflows, reducing manual task effort; managed data storage and retrieval on AWS DynamoDB and administered database infrastructure (users, roles, security patches).
- Analyzed production data in Python, SQL and Tableau to surface hiring patterns; built AI/ML-based predictive models (85% accuracy) for candidate stability that reduced hiring time 30%.
- Partnered with development and design teams to propose and prioritize product improvements backed by quantitative analysis and query-driven insights; defined KPIs and benchmarks with QA in Jira and Selenium, improving data quality and consistency 25%.

MakeSense Inc. — Texas (remote)

01/2022 – 05/2023

Data Analyst Intern

- Analyzed API performance data to identify latency issues and optimization opportunities; built dashboards and reports to monitor operational performance metrics.
- Designed A/B tests to measure the impact of API updates; ran data-quality assessments and validations to ensure accuracy and reliability.
- Built predictive models (logistic regression, decision trees, random forests); supported data pipeline design — databases, ETL pipelines and data warehouses.

EDUCATION

Master of Science, Bioinformatics and Management — George Mason University, Fairfax, VA

05/2023

Coursework: Project Management, Risk Management, Software Architecture, Finance; Python, SQL, R and Tableau.

Bachelor of Science, Biological Sciences — Delaware State University, Dover, DE

05/2021

Coursework: Biostatistics, Genetics, Molecular Biology, Biochemistry, Pharmacology, General and Organic Chemistry, Senior Capstone research.

ACADEMIC PROJECTS

Predicting Hospital Readmissions — ML model on EHR data, demographics, diagnosis codes and admission history to flag high 30-day readmission risk; evaluated on accuracy, precision, F1-score and ROC-AUC.

EHR Database Management System — Relational EHR schema (demographics, history, treatments, labs) with constraints and role-based access control; HL7 interoperability review and a backup / disaster-recovery runbook.

Chronic Kidney Disease Prediction — Data-mining / ML model identifying individuals at risk of CKD; hyperparameters tuned with grid and random search for integration into a healthcare application.